

# Foresight-ML Deployment Phase Checklist — Group 22

Date: April 10, 2026 Verified by: Nandana Pradeep Branch: `main` (latest pull)

## A. Serving Layer — Harshit + Bryan

| #   | Task                                                                                                                                                                                                                   | Status        | Evidence                                                                             |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------------------------------------------------------------------------------------|
| A1  | FastAPI service with 6 endpoints<br>( <code>/predict</code> ), ( <code>/company/{cik}</code> ),<br>( <code>/alerts</code> ), ( <code>/health</code> ),<br>( <code>/model/info</code> ), ( <code>/drift/status</code> ) | ✓<br>Done     | ( <code>src/api/main.py</code> ) + 5 routers in<br>( <code>src/api/routers/</code> ) |
| A2  | Pydantic request/response schemas                                                                                                                                                                                      | ✓<br>Done     | ( <code>src/api/schemas.py</code> )                                                  |
| A3  | Model + scaler loading on startup                                                                                                                                                                                      | ⚠️<br>Partial | API returns healthy; real model loading needs verification                           |
| A4  | API key auth via Secret Manager                                                                                                                                                                                        | ✓<br>Done     | ( <code>src/api/dependencies.py</code> ) (expanded to 45+ lines)                     |
| A5  | Rate limiting (slowapi — 100 req/min)                                                                                                                                                                                  | ✓<br>Done     | Configured in ( <code>src/api/main.py</code> )                                       |
| A6  | CORS configuration                                                                                                                                                                                                     | ✓<br>Done     | Configured in ( <code>src/api/main.py</code> )                                       |
| A7  | ( <code>get_top_features()</code> ) helper in<br>( <code>explain.py</code> )                                                                                                                                           | ✓<br>Done     | ( <code>src/models/explain.py</code> ) updated                                       |
| A8  | ( <code>Dockerfile.api</code> )                                                                                                                                                                                        | ✓<br>Done     | ( <code>deployment/docker/Dockerfile.api</code> )                                    |
| A9  | ( <code>cd-api.yml</code> ) GitHub Actions workflow                                                                                                                                                                    | ✓<br>Done     | ( <code>.github/workflows/cd-api.yml</code> )                                        |
| A10 | Cloud Run Service provisioned<br>( <code>foresight-api</code> )                                                                                                                                                        | ✓<br>Done     | Live at URL, ( <code>/health</code> ) returns ( <code>{"status": "healthy"}</code> ) |

## B. Dashboard UI — Nandana

| #   | Task                                                                                          | Status | Evidence                                                        |
|-----|-----------------------------------------------------------------------------------------------|--------|-----------------------------------------------------------------|
| B1  | <code>app.py</code> — multi-page layout, sidebar navigation, loading splash                   | ✓ Done | <code>src/dashboard/app.py</code> (335 lines)                   |
| B2  | Page 1 — Company Risk Explorer (search, trend chart, SHAP top-5, signals, financial snapshot) | ✓ Done | <code>src/dashboard/pages/company_risk.py</code> (485 lines)    |
| B3  | Page 2 — High-Risk Watchlist (filters, CSV export, sector chart, trend indicators)            | ✓ Done | <code>src/dashboard/pages/watchlist.py</code> (291 lines)       |
| B4  | Page 3 — Model Health (model card, drift status, prediction histogram, slice table)           | ✓ Done | <code>src/dashboard/pages/model_health.py</code> (262 lines)    |
| B5  | Page 4 — Pipeline Status (task lists, status dots, summary metrics, artifact grid)            | ✓ Done | <code>src/dashboard/pages/pipeline_status.py</code> (262 lines) |
| B6  | GCS data loader with caching                                                                  | ✓ Done | <code>src/dashboard/data/gcs_loader.py</code> (290 lines)       |
| B7  | API client with graceful fallback                                                             | ✓ Done | <code>src/dashboard/data/api_client.py</code> (86 lines)        |
| B8  | Shared utilities (risk helpers, formatting, chart theme)                                      | ✓ Done | <code>src/dashboard/utils.py</code> (160 lines)                 |
| B9  | Dashboard tests                                                                               | ✓ Done | <code>tests/test_dashboard.py</code> (1124 lines)               |
| B10 | Dashboard README                                                                              | ✓ Done | <code>src/dashboard/README.md</code> (188 lines)                |
| B11 | <code>Dockerfile.dashboard</code>                                                             | ✓ Done | <code>deployment/docker/Dockerfile.dashboard</code>             |
| B12 | <code>cd-dashboard.yml</code> GitHub Actions workflow                                         | ✓ Done | <code>.github/workflows/cd-dashboard.yml</code>                 |

| #   | Task                            | Status        | Evidence                      |
|-----|---------------------------------|---------------|-------------------------------|
| B13 | Cloud Run Service for dashboard | ⚠️<br>Unknown | Need to verify deployment URL |

## C. Drift Monitoring + Retraining Trigger — PiriyaJeishree

| #  | Task                                                          | Status        | Evidence                                                |
|----|---------------------------------------------------------------|---------------|---------------------------------------------------------|
| C1 | Data drift monitoring (Evidently AI)                          | ✅ Done        | <code>monitoring/data_drift.py</code>                   |
| C2 | Model drift detection                                         | ✅ Done        | <code>monitoring/model_drift.py</code>                  |
| C3 | Performance metrics monitoring                                | ✅ Done        | <code>monitoring/metrics.py</code>                      |
| C4 | <code>retrain_trigger.py</code> — reads retrain flag from GCS | ❌<br>Missing  | File not found in repo                                  |
| C5 | Airflow DAG extended (drift → branch → retrain/skip)          | ⚠️<br>Partial | DAG files updated but retrain branch needs verification |

## D. Model Versioning + Inference Schema — Palak

| #  | Task                                                                                                                                                                                          | Status    | Evidence                                                    |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------------------------------------------------------------|
| D1 | Versioning columns in <code>scores.parquet</code> ( <code>model_version</code> , <code>mlflow_run_id</code> , <code>trained_at</code> , <code>scored_at</code> , <code>model_roc_auc</code> ) | ✅<br>Done | <code>src/models/predict.py</code> updated                  |
| D2 | <code>manifest.json</code> write logic alongside <code>scores.parquet</code>                                                                                                                  | ✅<br>Done | <code>src/models/manifest_schema.py</code> updated          |
| D3 | Rollback event writes <code>rollback_triggered.json</code>                                                                                                                                    | ✅<br>Done | <code>src/models/registry.py</code> (expanded to 145 lines) |

## E. CI/CD Hardening — PiriyaJeishree

| #  | Task                                                           | Status        | Evidence                                                                     |
|----|----------------------------------------------------------------|---------------|------------------------------------------------------------------------------|
| E1 | SonarCloud integration + <code>sonar-project.properties</code> | ❌ Missing     | No <code>sonar-project.properties</code> found                               |
| E2 | Trivy container scanning in CD workflows                       | ⚠️ Unverified | Workflows updated but Trivy steps need confirmation                          |
| E3 | Slack notifications in CD workflows                            | ✅ Done        | <code>deployment/slack_adapter/</code> + <code>infra/slack_adapter.tf</code> |
| E4 | Pre-commit hooks ( <code>.pre-commit-config.yaml</code> )      | ✅ Done        | Ruff, ruff-format, mypy, gitleaks, Terraform configured                      |
| E5 | Branch protection rules on <code>main</code>                   | ⚠️ Unknown    | Cannot verify from CLI; check GitHub repo settings                           |

## F. Observability — Bryan

| #  | Task                                                          | Status | Evidence                                                |
|----|---------------------------------------------------------------|--------|---------------------------------------------------------|
| F1 | Structured JSON logging ( <code>src/utils/logging.py</code> ) | ✅ Done | <code>StructuredFormatter</code> class with JSON output |
| F2 | Cloud Monitoring alert: Cloud Run job failure                 | ✅ Done | <code>infra/monitoring.tf</code> (204 lines, expanded)  |
| F3 | Cloud Monitoring alert: ROC-AUC < 0.85                        | ✅ Done | <code>infra/monitoring.tf</code>                        |
| F4 | Cloud Monitoring alert: drift detected                        | ✅ Done | <code>infra/monitoring.tf</code>                        |
| F5 | Alert smoke test script                                       | ✅ Done | <code>scripts/cloud_run_alert_smoke_test.sh</code>      |

## G. Infrastructure — Bryan

| #  | Task                                 | Status | Evidence                                |
|----|--------------------------------------|--------|-----------------------------------------|
| G1 | Terraform: Cloud Run for API         | ✓ Done | <code>infra/api_service.tf</code>       |
| G2 | Terraform: Cloud Run for dashboard   | ✓ Done | <code>infra/dashboard_service.tf</code> |
| G3 | Terraform: Monitoring alert policies | ✓ Done | <code>infra/monitoring.tf</code>        |
| G4 | IAM: API service account permissions | ✓ Done | <code>infra/iam.tf</code>               |
| G5 | GCP Secret Manager                   | ✓ Done | <code>infra/secret_manager.tf</code>    |
| G6 | Artifact Registry                    | ✓ Done | <code>infra/artifact_registry.tf</code> |

## H. Testing

| #  | Task                                 | Status    | Evidence                                                                                                                             |
|----|--------------------------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------|
| H1 | Dashboard unit tests                 | ✓ Done    | <code>tests/test_dashboard.py</code> (1124 lines)                                                                                    |
| H2 | Integration tests: API e2e           | ✓ Done    | <code>tests/integration/test_api_e2e.py</code> (138 lines)                                                                           |
| H3 | API unit tests                       | ✓ Done    | <code>tests/test_api/test_predict.py</code> , <code>test_alerts.py</code> , <code>test_health.py</code> , <code>test_logic.py</code> |
| H4 | Registry tests (rollback + manifest) | ✓ Done    | <code>tests/test_registry.py</code> (expanded)                                                                                       |
| H5 | Integration test: drift monitor      | ✗ Missing | No <code>test_drift_monitor.py</code> found                                                                                          |
| H6 | All other existing tests             | ✓ Done    | 153 original tests still passing                                                                                                     |

I. Documentation + Demo

| #  | Task                               | Status        | Evidence                                                              |
|----|------------------------------------|---------------|-----------------------------------------------------------------------|
| I1 | Dashboard README                   | ✔ Done        | <code>src/dashboard/README.md</code>                                  |
| I2 | Main README updated                | ✔ Done        | <code>README.md</code> updated in latest pull                         |
| I3 | Notebooks: model experiments       | ✔ Done        | <code>notebooks/model_experiments.ipynb</code> (expanded)             |
| I4 | Notebooks: distress label analysis | ✔ Done        | <code>notebooks/distress_label_analysis.ipynb</code> (new, 685 lines) |
| I5 | Demo video (8-10 min)              | ✘ Not started | Nandana leads                                                         |

Summary

| Category                | Done | Partial/Unknown | Missing |
|-------------------------|------|-----------------|---------|
| A. Serving Layer        | 9    | 1               | 0       |
| B. Dashboard UI         | 12   | 1               | 0       |
| C. Drift Monitoring     | 3    | 1               | 1       |
| D. Model Versioning     | 3    | 0               | 0       |
| E. CI/CD Hardening      | 2    | 2               | 1       |
| F. Observability        | 5    | 0               | 0       |
| G. Infrastructure       | 6    | 0               | 0       |
| H. Testing              | 4    | 0               | 1       |
| I. Documentation + Demo | 4    | 0               | 1       |
| Total                   | 48   | 5               | 4       |

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## Action Items — Missing / Needs Attention

| # | Item                                                                     | Owner                          | Priority |
|---|--------------------------------------------------------------------------|--------------------------------|----------|
| 1 | <code>retrain_trigger.py</code> — not found in repo                      | Piriyajeishree                 | High     |
| 2 | <code>sonar-project.properties</code> — SonarCloud not configured        | Piriyajeishree                 | Medium   |
| 3 | <code>test_drift_monitor.py</code> — integration test missing            | Bryan                          | Medium   |
| 4 | Dashboard Cloud Run deployment — verify URL                              | Bryan                          | Medium   |
| 5 | Trivy steps in workflow files — verify presence                          | Piriyajeishree                 | Low      |
| 6 | Branch protection rules on <code>main</code> — verify in GitHub settings | Piriyajeishree                 | Low      |
| 7 | Demo video (8-10 min) — plan, record, edit                               | Nandana                        | High     |
| 8 | GCP billing account disabled — 403 errors on GCS                         | Team<br>(Bryan/Piriyajeishree) | High     |